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2023 PURCHASING STRATEGY REPORT

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Purchasing and Supply Chain Strategy for the new

"SAMSUNG GALAXY P1 Pro"

1.0. Executive Summary

We as the purchasing team created a strategy for the launch of a new product in November 2023, selecting Samsung as the company and introducing the Samsung Galaxy P1 Pro to satisfy market demand. Furthermore, to be competitive, the purchasing team comprised some ethical and distinctive ways of carrying out the purchase and supply chain processes in this report. We did this by choosing distinctive pricing approaches for the product as well as forecasting demand to make it compatible with Samsung's annual sales revenue. Finally, the report covers product sourcing strategies, inventory control, and strategic focus, highlighting successful measures and upcoming metrics.

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2.0. Introduction

The report offers a thorough analysis and tactical suggestions for Samsung Galaxy P1 Pro supply chain and purchasing. As the members of the purchasing group, it is our goal to create a productive supply chain that will enable the new product's successful launch and continuous manufacturing. This paper explains critical issues and offers practical insights to guarantee a smooth and economical supply chain for Samsung, with an emphasis on optimizing sourcing, supplier management, as well as transportation.

3.0. Background

Samsung is preparing to introduce a revolutionary product, focusing on a comprehensive supply chain, and purchasing strategy to ensure its successful launch and ongoing manufacturing. The goal is to ensure the entire supply chain operates effectively and efficiently, meeting product requirements throughout its lifespan. Moreover, Samsung's new product, featuring cutting-edge technology, novel features, and a unique value proposition, is a significant milestone for the company. Its focus on quality, performance, and user experience is crucial for Samsung's growth and market position. As a result, this paper aims to provide a comprehensive supply and purchasing chain plan for Samsung's new product, focusing on durability, economy, and customer-focused supply chain. Utilizing research insights, Samsung can achieve market dominance and achieve overall success.

4.0. Procurement and Supply Chain Strategy

4.1. Strategic Analysis

Company Vision

Samsung's vision is to transform lives through cutting-edge technologies, goods, and services, leading the digital age and promoting sustainable prosperity. With enthusiasm for innovation and commitment to perfection, they aim to revolutionize industries and liberate people.

Company Mission

The mission of Samsung is that Samsung aims to revolutionize global lifestyles by providing high-quality products and services that integrate technology into daily life. They strive to be a respected, trusted firm, promoting honesty, collaboration, and innovation, fostering global development and well-being.

History of Samsung

Samsung's history is marked by aspiration, invention, and determination. From humble beginnings in South Korea to becoming a global leader in technology, the company has transformed industries and reshaped people's lives through excellence and innovation. According to, (Chung, 2016) Samsung, Korea's largest corporate organization, expanded rapidly under the leadership of second-generation Chairman Kun-hee

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Lee. Sales increased from 10 trillion Won in 1987 to 350 trillion Won in 2014, accounting for 28% of Korea's total exports.

Moreover, as stated by (Chung, 2016) Samsung Electronics, a subsidiary of Samsung, produces and sells parts like transistors and display panels. The company's growth was facilitated by management innovations and cooperative reorganization. Samsung's SCM innovation strategy led to its ranking ninth in 2015 by Gartner among Fortune 500 and Forbes 2000 companies. This study examines how Samsung applied TQM principles to achieve such breakthroughs in SCM.

Furthermore, (Chung, 2016) shows that, Samsung used to duplicate popular Japanese products to create identical goods. However, it now focuses on analyzing rival products and the manufacturing process, examining components to meet consumer requirements in the target market.

Product range and market position

Technology has transformed our interactions and time management, with Samsung demonstrating innovation and setting new benchmarks in the IT sector. As a global consumer electronics innovator, they offer diverse products. According to, (YU, 2023) Samsung's product line expands from consumer electronics to complex technology, with an average annual growth of 38%, adapting to global trends. Moreover, (Omer, 2019) The following product categories are offered by Samsung.

- Tablets
- Cell phones, including smart phones and standard phones.
- LED and LCD televisions.
- HDTV, SMART TV, plasma TV, etc.
- Cameras and video cameras
- refrigerators
- Refrigerators
- Laundry machine
- The use of microwaves
- Laptops, printers, and accessories for IT.

Market positioning involves strategically creating a unique image, identity, and perception of a company, brand, or product, defining, and communicating its unique value and advantages. According to, (TomaszSzot1, 2019) Samsung dominates the global smartphone market, with flagship devices like Galaxy S and Note gaining popularity due to its well-known brand, diverse product range, and advanced features.

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Samsung dominates the television market with various models featuring QLED and Micro LED display technologies, maintaining a competitive edge through high quality, technological advancements, and user-friendly features.

Samsung dominates the home equipment market with energy-efficient products, advanced technology, and innovative designs, gaining market share through Galaxy series.

Competitors

Samsung faces competition from various industries, with major smart pen competitors primarily targeting consumer electronics. Therefore, the major competitors that Samsung should consider are.

1.0. **LG** and Samsung compete for market share as well as technological developments in consumer products, televisions, appliances, including audio systems.
2.0. In addition, In the field of consumer electronics**, Sony** is another well-known rival, particularly in the fields of televisions, audio equipment, as well as gaming consoles.
3.0. In several consumer electronics categories, like televisions, home appliances, including audio systems, **Panasonic** competes with Samsung.

Moreover, Samsung faces competition from **Apple, Huawei, Xiaomi**, and **Intel** in mobile devices, semiconductors, and electronic components.

How is the product positioned within the company's product range and the target market?

When positioning our product which is **"Samsung galaxy P1 Pro"** There are several features that has been considered when positioning the product in Samsung's product range. Below are several ways the product is being positioned within the company's product range.

1.0. Identify key characteristics and skills that differentiate digital-smart pens from competitors, including pressure sensitivity, accurate handwriting recognition, excellent Samsung device connection, and innovative functions.
    - Analyzing Samsung's brand identity, focusing on digital smart pen's advantages, convenience, creativity, productivity, and reputation for high-quality products and innovation.
    - In addition, considering the digital-smart pen's smooth integration with current Samsung gadgets, such mobile devices, tablets, or smart notebooks. Showcase how utilizing a pen with Samsung's network of goods improves user experience and efficiency.

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- Providing customers with assistance, tutorials, and instructional materials to help them get most of the digital smart pen. This can contain tutorials, user manuals, and online forums where users can exchange advice.

Samsung, a South Korean conglomerate, dominates consumer electronics, mobile technology, and corporate solutions by targeting diverse markets with products and services.

The digital smart pen from Samsung targets specific customer groups and must achieve its intended market placement.

- Samsung can promote digital smart pens as essential tools for **teachers and students**, offering features like notetaking, handwriting recognition, and compatibility with educational platforms for enhanced collaboration.
- Samsung's digital smart pen targets **professionals** in various industries, offering seamless productivity tools, realistic drawing capabilities, and simple information sharing. Emphasizing productivity, streamlining processes, and innovation could attract a diverse audience.
- Samsung's smart pen targets **artists and designers** with features like pressure sensitivity, precise tracking, and art software compatibility, allowing easy design tweaking, production, and sharing across various devices.
- Intelligent pen attracts **technology enthusiasts** seeking a seamless user experience with Samsung devices, showcasing innovative networking, gesture control, and technical innovations.
- Samsung's new smart pen offers unique camera capabilities and recording features, targeting **travelers and tourists**, offering a standout advantage over competitors.

These are the main focusing areas of when considering the target market for the new smart pen which is **"Samsung Galaxy p1 Pro."**

Product pricing aligned with competitors.

Samsung employs two main pricing strategies: value-based pricing and penetration pricing. For its new Galaxy P1 pro, Samsung uses market penetration/destroyer pricing strategy to align with competitors. This strategy aims to establish products at lower prices, attracting customers, creating demand, and fostering brand loyalty. This approach helps Samsung gain market share and a solid position in the industry.

Samsung plans to launch the Galaxy P1 Pro at $199, targeting high-end digital pens starting at $150 or several hundred dollars. The product features double cameras, a high-quality display, and long battery life. Samsung's finance team considers psychological pricing to make it unique.

Components within the product

Samsung Galaxy P1 Pro is a multi-functional digital pen with a unique preposition, offering a standout advantage in the market with **15 sub-components where two of them are features incorporated in the device.**

The **7-inch pen** features ballpoint ink, **Android 13 operating system**, **One UI 6.0**, and **IP68** certification, providing a powerful, entertaining experience even underwater.

Moreover, the pen is designed to be used according to users' convenience and the way they prefer. Samsung's **InkMate** Application allows users to customize their preferred themes for free, allowing for in-built designs and free installation.

InkMate simplifies pen control and handling with customizable themes, file updates, device power, and camera features, allowing users to capture photos, record audio, and adjust speaker sound.

(Refer appendix A)

Appendix A: Sub-Components of the product .....

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Supply chain diagram

Suppliers

NXP Semiconductors | Nikon | Seoul Semiconductors | Thor labs | OpenX cell | Qualcomm | Mitsubishi Pencil Company | AAC Technologies | Vesper Technologies | Tisco steel group | Molex Illinois | DNA Packaging systems

Above-mentioned suppliers provide sub-components, which includes, **Chipset, Camera, Torch, Laser beam, App, Biometrics, Rubber grip, speakers, Microphone, Stainless steel outer coverage Water-resistant** and **packaging** feature to the Samsung-south Korea at GUMI complex.

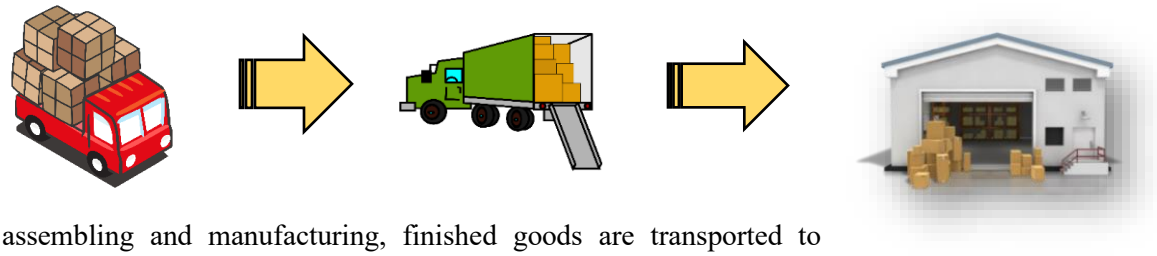
Manufacturer

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Samsung South Korea's GUMI complex will handle manufacturing, assembling sub-components like OS, battery, storage, multi-color ink, and display for the Galaxy P1 pro. The finished goods will be delivered to South Korean warehouses, followed by manufacturing in other countries.

Distribution

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After assembling and manufacturing, finished goods are transported to Samsung's Gwangju distribution center in South Korea. After distributing to other countries, finished products are warehoused and distributed to retailers.

Retailers

Major retailers in South Korea

emart LOTTEMart Home plus coupang Gmarket

Samsung plans to distribute sample stock to major retailers in South Korea, delivering future stocks to these retailers and other countries, including other major retailers.

Retailers in USA, Australia, and India

Walmart amazon bing JB HI-FI Flipkart RELIANCE

Customers



Finally, from the retailers, the **"Samsung Galaxy P1 Pro"** is delivered to the final consumer through the retail outlets of Samsung. Thereby, after the sample stock, the same process will be continued.

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Figure 1: Supply Chain Diagram

Primary significant competitors for the product

According to, (Mohamad Wehbi 1, 2022) Smart pens like Apple Pencil and Samsung S pen are mobile device attachments, limited to tablets. Apple's Apple Pencil offers a comfortable handwriting experience, making it a major competitor to Samsung.

Microsoft's Surface Pen dominates the digital pens market with precise inking, pressure sensitivity, and tilting recognition, seamlessly integrated with Surface devices.

In addition, **Wacom** is a well-known company that dominates the digital pen market and is well-known for its high-quality stylus products. According to (Pacholec & Nitra, 2011) The Wacom Tablet PL comes with a digital pen that may be utilized for writing on the screen in addition to replacing the mouse.

HP digital pens are compatible with touch-enabled products, making them a major competitor for Samsung Galaxy P1 Pro.

Company's Purchasing and Supply Chain Competitive advantages.

Samsung can gain a competitive advantage in the purchasing and supply chain of the Galaxy P1 Pro by fostering strong supplier relationships. By maintaining communication, ensuring rapid delivery, quality control, and continuous improvement, Samsung can maximize its supply chain and increase overall efficiency.

Samsung utilizes Lean manufacturing and logistics for sustainable and ethical sourcing, enhancing its product supply chain. By reducing waste, optimizing processes, and managing just-in-time inventory, the company can produce pens faster, meet consumer demands, and reduce surplus inventory.

Integrating vertically in Samsung's pen manufacturing process provides a competitive edge by achieving internal production stages, controlling component availability, cost, and quality, and reducing dependency on external suppliers.

4.2. Sourcing Strategy

Types of Sourcing plans employed.

The raw materials needed to manufacture "Galaxy P1 Pro", are sourced from different geographic regions. All the materials are being collected from the Port of Incheon, the main port in South Korea. Although most of the Samsung products are made in Vietnam, "Galaxy P1 Pro" would be made initially in South Korea, and later the production operations would be reallocated to the Vietnam, the global manufacturing hub of Samsung.

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Global sourcing strategy is chosen for 90% of the procurement process. This is because it would provide a greater financial edge for the global manufacturers as well as for Samsung. It also contributes to cost – efficient labor, export incentives, beneficial regulations, and possibility to acquire the most favorable and quality infrastructure. In Addition, global sourcing will result in Preferential Trade Agreements that will facilitate in sustainable supplier relationships in long-term (Jain, et al., 2014). Furthermore, as our sourcing needs are broad, we have acquired a responsive sourcing strategy too. Responsive sourcing strategy is also known as sourcing from domestic organisations. This method was selected because, it is convenient to conduct the market analysis more effectively and the delivery lead time is much shorter (Wu & Zhang, May, 2014).

A crucial move of the sourcing plan was to apprehend the Sourcing needs of the newly designed product. As mentioned above, both the domestic and international sourcing options were considered. However, greater weight was given to global sourcing strategy. Moreover, most of the subcomponents of the product is sourced strategically, which denotes the alignment of business strategies with the purchasing strategies leading to overall business performance (Formentini, et al., 2019).

The Galaxy P1 Pro features UFS 3.1 storage technology, manufactured by SAMSUNG, which will be strategically sourced and controlled by the company. The battery and charging pad are made by SAMSUNG SDI, a subsidiary of the company. Seoul Semiconductor, a leading LED manufacturer, will supply the torch, which will be sourced domestically for reliability, approachability, and reputation. The display screen and theme-changing function will also be built by SAMSUNG.

The Galaxy P1 Pro is a Samsung pen with Android operating system, sourced from Canadian company The Pen Counter, Vesper Technologies, AAC Technologies, Tisco Steel Group, Nikon, Mitsubishi, and NXP Semiconductors. It features a built-in camera, rubber grip, fingerprint sensor, infrared IR laser beam, water-resistant feature, and 32-bit RISC ARM-based Cortex-M7 chipset. The app allows users to control features, themes, and brightness.

Issues that should be addressed

The purchasing team faces issues in sourcing subcomponents, particularly in selecting the preferred supplier. Vesper Technologies was chosen due to its exceptional resistance to dust particles, water, and shock, making it suitable for challenging environmental settings. Suppliers are chosen after extensive research and evaluation to meet product standards.

Another challenge that was encountered was global sourcing. Even though global sourcing facilitates many benefits, it may be complicated at times. Some instances are the regional, legal, and cultural boundaries that

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must be addressed, also it is crucial to adhere to the global trade regulations and standards, which will eventually bring mismanagements in the Supply Chain. Furthermore, difficulties may arise due to the quality complexities that may arise when dealing with suppliers beyond borders. For instance, when the chipset of "Galaxy P1 Pro" is considered, there were initially two potential suppliers were highlighted in the list of suppliers, that is, DH electronics in Germany and NXP Semiconductors in Netherlands, out of which NXP Semiconductors company was selected after inspecting the quality attributes and standardization of such a sensitive subcomponent and by effective coordination between the suppliers (Olivares Tenorio, et al., 2021).

Another difficulty confronted by our team when sourcing was the exchange rates. Although domestic sourcing was able to execute smoothly due to stable inflation, when sourcing internationally, the exchange rates played a huge role, mainly because, to determine the exact cost for the inputs and other production costs, to gauge the profitability and revenues as currency fluctuations may influence the financial benefits that the product may bring. Thus, when sourcing for "Galaxy P1 Pro, the below mentioned exchange rates were considered.

- 1 USD = 1,316.30 South Korean won
- 1 Indian Rupee = 16.03 South Korean won
- 1 Canadian Dollar = 993.96 South Korean won
- 1 Chinese yuan = 181.50 South Korean won
- 1 Japanese Yen= 9.12 South Korean won
- 1 Euro equals= 1,436.10 South Korean won

Supplier Relationships

To gain financial and operational benefits, as the purchasing team, we believe entering a sustainable, long-term relationship with the supplier would be ideal most of the time, because the commitment of the manufacturer to the product is a vital factor (Chen, et al., 2023). But with the broadening of the production plans, it would vary.

The below table illustrates the type of the relationship with each supplier of the subcomponents (It is important to note that all the relationship purchasing is single sourced, to have operational cost benefits and all the tactical purchasing are dual sourced, to reduce the risks associated with the subcomponents)

(Refer Appendix B)

Appendix B: Supplier Relationships.....

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Transporting Mechanisms

In the past few years, the need for effective and efficient Supply Chain Management was recognized by SAMSUNG as it will enhance the company's operations including planning and executing functions. Also, SAMSUNG has adopted a systematic Supply Chain procedures along with the concept called "Quality Movement", that focused on assuring the quality of the product as well as the business processes (Yang, et al., 2007).

The Galaxy P1 Pro is transported using Ocean Freight due to its cost-effectiveness, capacity for larger quantities, and environmental impact. Authorized sellers distribute the product to customers, and back-up products for repairs are also sent via Ocean Freight.

4.3. Strategy Implementation

4.3.1. Forecasting

Based on market analysis, historical data about digital pens, and input from sales and marketing teams provided within the financial statements and Earnings Releases Documents by Samsung through their main websites,

Demand Forecast for Samsung Smart Pen:

Market Size and Growth Rate:

The global digital pen market size was valued at USD 2.04 billion in 2021.

The market is expected to expand at a compound annual growth rate (CAGR) of 13.3% from 2022 to 2030.

Customer Preferences:

Based on the customer preferences we discussed earlier, we found that customers value the smart pen's precision, accuracy, and ability to work without the need for batteries or recharging.

Customers also appreciate the integration of smart pens with e-learning platforms, making them more convenient for students of all ages.

Moreover, the demand for smart pens has increased in sectors like banking and healthcare due to the need for error-proof database collection and digitization.

Product Insights:

Handwriting pens accounted for more than 70.0% of the market share in 2021.

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Assumption: Samsung's smart pen falls under the handwriting pen category.

Usage Insights:

Tablets occupied the highest market share of 58.6% in 2021, and the segment is projected to have the highest CAGR of 13.8% from 2022 to 2030.

Since the Samsung smart pen is designed to work with tablets and smartphones, we can expect significant demand from this usage segment.

Regional Insights:

The Asia Pacific is expected to be the fastest-growing market, with a CAGR of 14.3% from 2021 to 2030.

Assumption: Since Samsung is a Korean company located in Seoul, we can assume that it will have a fast-growing demand within the whole country and Asia.

Seasonal Variations:

Assumption: Demand due to decreased activity in companies and educational institutions over the summer vacation months (months 7 to 10), demand growth might slow down.

(Maintaining the same units sold from the previous month)

Demand Forecast for Galaxy P1 Pro Smart Pen:

To create the demand forecast for the Samsung smart pen in units for the next 12 months, we will use the estimated revenue growth rate and the average selling price of 250,000 KW.

Calculation for the estimated revenue in 2021:

Estimated revenue in 2021 = Market size in 2021 x Market share of handwriting pens

Estimated revenue in 2021 = USD 2.04 billion * 70% = USD 1.428 billion ≈ USD 1.43 billion

Based on the team's analysis and forecasting efforts, the following demand forecast and sales forecast in Korean Won has been developed for the Samsung Smart Pen over the next 12 months:

(Refer Appendix C)

Appendix C: Forecasting Calculations

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Month	Demand Forecast in units	Forecasted Sales Revenue (KW=Korean Won) (Billion)
4Q23 (Fourth quarter of 2023)		
November	6,480	1.62
December	7,360	1.84
1Q24 (First Quarter of 2024)		
January	8,360	2.09
February	9,440	2.36
March	10,640	2.66
2Q24 (Second Quarter of 2024)		
April	11,960	2.99
May	11,960	2.99
June	11,960	2.99
3Q24 (Third Quarter of 2024)		
July	11,960	2.99
August	11,960	2.99
September	13,560	3.39
4Q24 (Fourth Quarter of 2024)		
October	15,360	3.84

Table 1: Demand Forecasting

A forecast is a crucial tool for anticipating future demand and planning business operations accordingly. It provides valuable insights into customer needs, allows for effective production planning, resource allocation, and inventory management. By utilizing a forecast, Samsung Galaxy P1 Pro can align its supply chain, optimize inventory levels, and ensure timely availability of the Smart Pen to meet customer demand.

To ensure the accuracy and reliability of the forecast, the team employed a combination of qualitative and quantitative forecasting methods. Qualitative methods, such as market research on Samsung, which was complemented by quantitative methods, including time series analysis and regression analysis, which utilized historical sales data and profit/ Income Information reading the products of Samsung.

The team analyzed historical sales data and market trends to identify seasonal variations in demand for smart pens. They identified back-to-work seasons and holidays as high demand periods. Low demand during winter break for educational institutions may decrease demand. Budget-conscious consumers may prioritize essential purchases over discretionary spending.

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Moreover, the team considered the different stages of the Smart Pen's product lifecycle when developing the forecast. From the initial introduction phase to the subsequent growth, maturity, and potential decline phases, each stage presents unique challenges and opportunities. The team anticipated demand fluctuations based on factors such as product launches, customer adoption rates, technological advancements, and market saturation, pricing strategies and promotional activities.

With the team assumptions the Smart Pen might reach the maturity phase of its product lifecycle during the months of May to August where it may experience a decline in demand. The initial excitement and novelty factor may have worn off, leading to a natural decrease in demand while the seasonal effects before potential marketing or product updates in the upcoming months.

Anyhow by aligning production and inventory management strategies with the specific demands of each lifecycle stage, Samsung can optimize resource allocation, minimize costs, and ensure the Smart Pen remains competitive throughout its lifecycle.

Samsung's smart pen production strategy relies on forecasts for informed decisions, procurement, and planning. Forecasts ensure timely availability of raw materials, optimize manufacturing processes, and allocate resources efficiently, minimizing inventory carrying costs and stockouts.

4.3.2. Systems

Systems that will be employed to manage purchasing and supply chain activities.

In their quest for supply chain excellence, we have come up with management systems to embrace a cutting-edge ecosystem of interconnected systems to orchestrate the purchasing and supply chain activities for the innovative Samsung Galaxy P1 Pro Smart Pen.

ERP (Enterprise Resource Planning).

ERP system that integrates key business processes, including order management, inventory control, production planning, finance, procurement, human resources, and customer relationship management which serves as the SCM's skeleton by acting as a central database and by supplying and monitoring real-time information. (Leung, 2016)

Samsung utilizes ERP systems like SAP ERP, S/4HANA, Fiori, and Ariba for HEC, application, and infrastructure support, and plans to integrate these systems into their SCM systems.

ERP system will handle:

• Administering financial transactions, budgeting, and financial reporting relating to the supply chain.

• Simplifying the process of buying the components and raw materials required for manufacturing.

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• Managing inventory replenishment, monitoring stock movements, and optimizing inventory levels.

• Managing training, performance reviews, and workforce statistics for SCM workers.

Benefits of integrating ERP into Supply Chain Management:

• Automating repetitive processes lowers human error rates and increases process effectiveness.

• Better financial visibility and accurate reporting result in more informed financial decisions, which improves financial control.

• Effective resource allocation makes sure that assets are used effectively and cuts down on waste.

• Maintain optimal inventory levels to reduce carrying costs and guarantee on-time order fulfillment.

• Enhance cost management and financial forecasts to raise total supply chain profitability.

Targets to achieve:

- Reduce Smart Pen procurement lead times by 10% through streamlining purchasing processes.
- Optimize inventory levels to minimize carrying costs.
- Improve financial forecasting and cost control to enhance overall profitability.

OMS (Order Management System).

An OMS software system automates and streamlines the order fulfillment process, from order capture to delivery. It acts as the nerve center of the supply chain, seamlessly connecting the SAP ERP and SCM systems where customers can effortlessly place orders for the Samsung Galaxy P1 Pro Smart Pen, receive instant updates on order status, and track their shipments with ease.

The OMS will handle:

- Receiving and processing customer orders for Smart Pens.
- Providing real-time updates on order status and shipment tracking.
- Handling order confirmations, shipping notifications, and customer inquiries.

Benefits of integrating OMS:

- Streamlined order processing leads to faster delivery times and improved customer satisfaction.
- Customers can track their orders in real-time, reducing support inquiries and enhancing transparency. (Yoon, 2020)
- OMS data integration with ERP and SAP SCM systems ensures accurate order status updates.

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Targets to achieve:

- Enhance customer satisfaction through timely order fulfillment and transparent tracking.
- Achieve real-time visibility of Smart Pen orders, enabling proactive order tracking and customer communication.
- Enhance Smart Pen customer communication and satisfaction by providing accurate and timely order updates.
- Reduce Smart Pen returns processing time by 25% through efficient returns and exchange management.

APS (Advanced Planning and Scheduling)

Samsung already has integrated Advanced Planning and Scheduling (APS) system for the processes to automate and optimize the management of materials and production in its supply chain which will be continued in the supply chain and purchasing activities of the smart pen.

The APS system will handle:

- Demand Forecasting: Analyzing historical data and market trends to generate accurate Smart Pen demand forecasts. (Chen, 2017)
- Production Planning: Creating optimized production plans based on demand forecasts, resource availability, and constraints. (Li, 2017)
- Developing detailed production schedules, considering machine capacities, labor availability, and order priorities.
- Simulating different scenarios to evaluate the impact of changes in demand or resources on production plans.

Benefits of integrating APS:

1. Improved production planning leads to enhanced resource utilization and reduced bottlenecks.
2. Aligning production with demand reduces stockouts and improves customer satisfaction.
3. Allows to dynamically adjust production levels and schedules based on real-time demand fluctuations.

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Targets to achieve:

- Improve Smart Pen production efficiency and Inventory turnover ratio by 15% through optimizing resource allocation and inventory control.
- Enhance on-time delivery performance by aligning production plans with actual demand.
- Increase overall supply chain agility and responsiveness.

4.3.3. Inventory Policy

Inventory Policy and Stock Levels:

1. Just-in-Time (JIT) Inventory Policy:

JIT inventory policy for the Samsung Galaxy P1 Pro will ensure that all components are received just in time for production, minimizing inventory levels. As soon as a customer place an order for the Smart Pen, the supply chain system immediately triggers the procurement of the required components and get it delivered to the production line allowing seamless assembly, and reduced need for excess stock and meeting customer demand efficiently while minimizing inventory holding costs.

2. Economic Order Quantity (EOQ) Inventory Policy:

Adopting the EOQ inventory policy helps optimize the order quantity for the Samsung Galaxy P1 Pro Smart Pen. Here, if the optimal order quantity is 500 Smart Pens, once the inventory level reaches the reorder point, the team initiates an order for another 500 which minimizes ordering costs and holding costs, ensuring efficient inventory management.

Buffer Stock for Smart Pen:

The buffer stock serves as a safety net, protecting against unexpected spikes in demand, production delays, or disruptions in the supply chain.

Buffer stock calculations considering the demand for each month:

(Refer Appendix D)

Appendix D: Buffer Stock Calculations.....

For the whole forecasted year, we will keep a total of 7,880 additional units of inventory just to be safe from risky situations in demand.

Independent Demand vs. Dependent Demand:

Independent demand refers to the Samsung Galaxy P1 Pro where we have decided to focus on forecasting customer demand based on historical data, market trends, and customer insights.

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Considering the dependent demand for components and raw materials required for manufacturing this, is required to employ production planning and supply chain coordination strategies or with our inventory. Samsung already employees the MRP (Material Requirement Planning) System which will help to calculate the required quantity of dependent items based on the production plan and the Bill of Materials. (BOM)

Ordering Information, Lead Times, and Delivery Methods/Frequencies:

Country	**Material**	**Lead Time**	**Order Frequency**	**Delivery Methods**
United states	Water Resistant	2 weeks	Monthly	International air freight International sea freight Domestic transportation
	Laser Beam	4 weeks	Every 3 months	
	Biometrics	3 weeks	Quarterly	
	Microphone	2 weeks	Monthly	
Japan	Camera	5 weeks	Every 3 months	
	Rubber Grip	2 weeks	Monthly	
China	Speakers	3 weeks	Every 3 Months	
	Stainless Steel	4 weeks	Quarterly	
Netherlands	Chipsets	3 weeks	Quarterly	
South Korea	Multi Color Components	2 weeks	Monthly	
	Storage Components	2 weeks	Monthly	
	Torch Components	3 weeks	Every 3 Months	
	Battery	2 weeks	Monthly	
	Display Screen	3 weeks	Quarterly	
Canada	Ballpoint Pen	2 weeks	Monthly	

Table 2: Deliver/order Information.

4.4. Strategic Direction for the product & Supply Chain Strategy

Outsourcing and Offshoring - To leverage outsourcing and offshoring strategies for the Galaxy P1 Pro, our team has strategically selected external suppliers for sourcing raw materials and components needed to manufacture the smart pen. Suppliers from different geographic regions are evaluated based on their cost

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advantages and specialized capabilities (Anand, 2000).Also the service of packaging is offshored from an Indian company (DNA Packaging). Though the production operations of the Galaxy P1 Pro are carried out in South Korea, leveraging the manufacturing capabilities of Samsung's main hub as part of the offshoring strategy, the production is reallocated to Vietnam, Samsung's global manufacturing hub, to benefit from cost-efficient labor, export incentives, and favorable infrastructure (Faisal, 2017). As the purchasing team, we intend to continue offshoring and outsourcing minor activities in the future.

Green and Sustainable Practices - As part of Samsung's commitment to sustainability and environmental responsibility, we incorporated green and sustainable practices of Eco-Friendly Packaging Optimization. We have opted to use wooden packaging which is a sustainable alternative to traditional packaging materials such as plastics or non-recyclable materials for the Galaxy P1 Pro smart pen. It is biodegradable, renewable, and helps reduce the carbon footprint associated with packaging. It helps to minimize excess space and material consumption, reducing both costs and environmental impact (Pagell, 2010).

Continuous Improvement and Kaizen - With Samsung's already implemented Lean Six Sigma principles, the team engages representatives from purchasing, production, logistics, and other relevant departments in cross-functional teams to embrace a culture of continuous improvement in the supply chain of the Galaxy P1 Pro. These teams work together to identify areas of improvement, streamline processes, and eliminate waste throughout the supply chain (Lamming, 2018) to systematically identify and eliminate non-value-added activities. The team continuously monitors key performance metrics and analyzes data to identify areas for improvement. They track metrics such as lead time, cycle time, throughput, and quality indicators to measure progress and identify opportunities for further Kaizen activities.

Kanban System Implementation - The team integrates suppliers into the Kanban system for seamless information flow and collaboration. They establish clear communication channels, share metrics, and use visual or digital systems for real-time visibility, identifying bottlenecks, and improving supply chain efficiency.

Blockchain Technology Integration - The Galaxy P1 Pro smart pen team uses blockchain to create secure, immutable records of transactions and interactions, ensuring traceability and authenticity. Smart contracts automate procurement processes, reducing manual paperwork and enhancing transactional efficiency.

To enhance the product performance, our team will furthermore develop 'Galaxy P1 Pro' by adding shock absorbers to improve the product durability. As the purchasing team, we intend to build strategic alliances with the supplier.

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As an attempt to improve the smart pen further, we intend in adding a feature that would allow the user to contact the nearest police station in an emergency.

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Appendices

Appendix A

Sub-Components	Features	Benefits
• **Stainless steel**	• The body of the pen is covered with a **stainless-steel pipe.** • The stainless-steel pipe is a combination of **stainless-steel sheets** to make it stronger. • The pen comes with elegant outer look with a glossy **Sapphire blue and Midnight blue**.	**1.** Helps to increase the **durability** of the pen. **2.** Gives **the premium, Professional and elegant look**. **3.** **Easiness to clean and maintain.**
• **Rubber Grip**	✓ Comes with rounded shape, **flexible rubber** which is not slippery because of **the tight dotted rubber surface**.	**4.** Comfortability when holding the pen. **5.** Helps to prevent Unintentional drops. **6.** **Lessens** the vibrations that the pen's internal parts transmit to the user's hand.
• **Infra-red laser beam**	✓ The **red laser beam** comes with **a LED bulb** inside to **erase** digitally or physically written texts. ✓ Can be seen indicating **red light** at the bottom of the pen. ✓ Includes the embedded sensor to automatically	• Decreased wasted paper and demand for disposable correctional materials. ✓ Real-time collaboration and continuous idea generation enabling easy content editing and amendment on screens and documents.

	detect when the user wants to erase something.	
• **Camera**	• It comes **with Nikon AF Fisheye-Nikkor 16mm f/2.8D** as the **primary camera** offering a 180-degree diagonal angle of view, producing a circular fish-eye effect on full-frame camera. • Pen includes a **2MP macro lens** as the **secondary camera** for text scanning and recording text related documents.	• Creates unique, creative images capturing small details that may be missed. • Benefiting travelers, photographers, and light-packers with a single device combines camera and digital pen. • **Versatility** which allows users to quickly switch from fisheye to macro lenses for landscapes and details.
• **Display**	✓ The display of the pen includes a **3-inch diagonal**-measured-long **Super AMOLED full HD display.** ✓ Comes with a Display Resolution of **800x480 pixels.**	• Helps to **provide crisp (**800x480 pixel) **graphics and texts.** enhancing readability and visual accuracy. • **Energy Savings** since the pen's battery lasts longer.
• **Chipset**	✓ The pen is powered by **32-bit RISC ARM-based Cortex – M7 chipset providing** higher **clock speed (**32MHz to 64MHz), more **advanced pipeline**	• Helps the digital pen to run **advanced programs, process images or recognize handwriting**. • Helpful in efficient **task completions** like handwriting detection,

	architecture, and a **larger cache.**	real-time input recognition, and complicated computations.
• **Biometrics**	✓ The pen includes biometrics such as **fingerprint scanning and voice recognition** required to unlock the device and to protect its recorded files and documents.	• **Eliminates time-consuming requirements** of passwords and PINs for secure authentication. • **Improved Security**. With fingerprint patterns and speech characteristics challenges unauthorized access.
• **Storage**	✓ The **storage capacity** is **512GB,** known as the **"UFS 3.1** that can deliver **sequential read speeds** of several GB per second (GB/s) or even **exceeding 3GB**.	✓ **Responsive storage capacity** since UFS 3.1 storage in digital pen offers compatibility with software, apps, file sizes. ✓ **Increased file transfer speed** enabling efficient content sharing between digital pens and computers.
• **Flashlight**	• The pens include a **flashlight** with a **bright LED bulb** located near the laser beam.	• Helps to **increase visibility** in writing and drawing. • Useful in **emergency situations** as a backup source for security, navigation, and

		signaling in emergencies.
• **Battery**	• The pen comes with **a long-life battery** capacity of **500mAh** powered by **a Li-ion battery.** • Includes a **5 watts wireless** charging pad. • Within**1 hour** period the user can fully charge the pen and use the device from **several days to a week of active usage.**	✓ **Offers an** extended battery life for uninterrupted writing and sketching. ✓ **Wireless Charging provides** convenience, simplicity, and reduces port deterioration.
• **Speakers**	• The pen consists of **micro speakers** embedded into the sides of the pen.	• **Enables** audio-based **note taking,** allowing users to merge written notes with related audio for comprehensive discussions and adaptability. • Enhances **pronunciation and listening comprehension.**
• **Microphone**	✓ The pen includes a built-in **mini microphone** near the charging coil.	• **Enables recording voice memos** for meetings, interviews, or presentations using this function, combining audio and notes for in-depth documentation.

		• **Voice Control and Command** enable users to interact with a pen, activate capabilities, navigate menus, and start features.
• **Color mechanism cartridge**	✓ The **removable and replaceable** color mechanism cartridge (Includes **Blue, Black, Red, Green, and Orange)** comes as **an internal process** which is being supported with the efficient operation of the chipset.	✓ **Improved creativity** enhancing visual appeal and engagement in drawings. ✓ **Visual divergence**. simplifying monitoring and identifying changes, additions, and annotations among contributors.

Appendix A: Sub-Components of the product

Appendix B

Subcomponent	Sourced Company	Type of Supplier Relationship	Reason
Ballpoint pen	The pen counter – Canada	Relationship Purchasing	Retaining the supplier for quality of ink supplies was the main intention.
Water resistant	Molex – Illinois, USA	Relationship Purchasing	The organisation engages in constant innovations, making room for "Galaxy P1 Pro" to improve further.
Multi colors	SAMSUNG - South Korea	Relationship Purchasing	Sustaining long-term relationships with raw material suppliers for in-house subcomponent manufacturing.

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Chipset	NXP Semiconductors, Eindhoven, Netherland	Relationship Purchasing	NXP Semiconductors selected for sensitive Galaxy P1 Pro subcomponent, aiming for long-term sustainability.
Storage	SAMSUNG - South Korea	Relationship Purchasing	SAMSUNG aims for sustainable, long-term relationships with suppliers for storage, resulting in financial benefits.
Torch	Seoul Semiconductors, South Korea	Relationship Purchasing	Considered factors such as Quality and accessibility, that align with the standards of SAMSUNG
Laser beam	Thor labs, New Jersey, USA	Relationship Purchasing	High potential subcomponent for Galaxy P1 pro success.
OS	Google	Relationship Purchasing	Implementing sustainable relationship purchasing strategy with Google for shared information.
Battery and Charging Pad	SAMSUNG SDI - South Korea	Relationship Purchasing	Long-term strategic alliances with battery component suppliers established through internal relationships with SAMSUNG subsidiary.
Speakers	AAC Technologies, China	Relationship Purchasing	SAMSUNG's collaboration with AAC Technologies is crucial for future innovations and maintaining the Galaxy P1 Pro's information.
Biometrics	Qualcomm, San Diego, USA	Relationship Purchasing	Since SAMSUNG has collaborated with AAC Technologies before, as the purchasing team of "Galaxy P1 Pro", we believe it is essential to sustain this company for future innovations as well

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			because of the information that has been shared with the company.
Rubber grip	Mitsubishi Pencil Company, Tokyo, Japan	Relationship Purchasing	Considered factors such as the quality and the reliability of the subcomponent and the possibility of innovations.
Display screen	SAMSUNG - South Korea	Relationship Purchasing	The suppliers of the components are filtered with a long-term orientation due to the availability of a significant amount of information
Microphone	Vesper Technologies, Inc, Boston, USA	Relationship Purchasing	Reason for having a close relationship with the company is because, it shows that potential of innovating and bringing performance and cost benefits for SAMSUNG
Camera	Nikon – Japan	Relationship Purchasing	Considered factors such as the quality and the reliability of the subcomponent and continuous innovations that would lead to improve the quality even more.
App	OpenXcell, Gujarat, India	Transactional Purchasing	Arms-length with suppliers is crucial due to limited information sharing and customer feedback-driven changes.
Packaging	DNA Packaging Systems - India	Relationship Purchasing	Because of the quality and the uniqueness of the package.
Stainless steel	Tisco steel group, China	Transactional Purchasing	SAMSUNG seeks strategic alliances with companies, including Tisco Steel for Galaxy P1 Pro, with long-term customer relationships.

Appendix B: Supplier Relationships

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Appendix C: Forecasting Calculations

Forecasting Calculations:

For Month 1:

- Estimated revenue for Month 1: USD 1.43 billion * (1 + 13.3%) = USD 1.62 billion

- Estimated units sold in Month 1: USD 1.62 billion / 250,000 KW = 6,480 units.

For Month 2:

- Estimated revenue for Month 2: USD 1.62 billion * (1 + 13.3%) = USD 1.84 billion

- Estimated units sold in Month 2: USD 1.84 billion / 250,000 KW = 7,360 units

Appendix D: Buffer Stock Calculations

Buffer Stock calculation:

4Q23: Buffer units required = 7,360 units - 6,480 units (November) = **880 units**

1Q24: Buffer units required = 10,640 units - 8,360 units (January) = **2,280 units**

2Q24: Buffer units required = 11,960 units (April, May, June - same demand for all three months) - 10,640 units (March) = **1,320 units**

3Q24: Buffer units required = 13,560 units - 11,960 units (June) = **1,600 units**

4Q24: Buffer units required = 15,360 units - 13,560 units (September) = **1,800 units**

Total buffer units required for all quarters: 880 + 2,280 + 1,320 + 1,600 + 1,800 = 7,880 units

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